

Kyle Hansen

Curriculum Vitae

Westmont College, Department of Mathematics
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Education

- June 2025 **Ph.D. in Mathematics**, *UC Santa Barbara (UCSB)*, Santa Barbara, CA
Advisor: Fedya Manin
Certificate in College and University Teaching (CCUT), *UCSB*, Santa Barbara, CA
- June 2021 **M.A. in Mathematics**, *UCSB*, Santa Barbara, CA
- May 2019 **B.S. in Mathematics**, *Westmont College*, Santa Barbara, CA
B.A. in Computer Science, *Westmont College*, Santa Barbara, CA

Employment

- 2025-Present **Assistant Professor**, *Mathematics*, Westmont College, Santa Barbara, CA
- 2021-2023 **Adjunct Professor**, *Mathematics*, Westmont College, Santa Barbara, CA
- 2023 **Adjunct Professor**, *Computer Science*, UCSB, Santa Barbara, CA
- 2022 **Lead Teaching Assistant**, *Mathematics*, UCSB, Santa Barbara, CA
- 2019-2025 **Teaching Assistant**, *Mathematics*, UCSB, Santa Barbara, CA

Teaching Experience

Instructor of Record

Discrete Mathematics, *Westmont College*
F 2025, F 2023, F 2022, F 2021

Abstract Alegbra, *Westmont College*
F 2025

Foundations of Computer Science, *UCSB*
Sp 2023

Upper Division Teaching Assistant

Linear Algebra, UCSB
Sp 2024, F 2023, W 2022*

Numerical Methods, UCSB
W 2024

Methods of Analysis, *UCSB*
F 2021, Su 2021*, W2025

Lower Division Teaching Assistant

Transition to Higher Mathematics, *UCSB*
Su 2024, Su 2023, W 2021*, Su 2020*, Sp 2020*

Linear Algebra with Applications, *UCSB*
F 2022, Su 2022, F 2020*

Differential Equations, *UCSB*
Sp 2021*

* this term was conducted remotely

Calculus for the Social and Life Sciences I, UCSB

W 2020

Calculus with Applications I, UCSB

F 2019

[Course Grader](#)

Modern Euclidean and Noneuclidean Geometry, UCSB

F 2024

Discrete Mathematics, Westmont College

F 2018, F 2017, F 2016

MA-020 Linear Algebra, Westmont College

Sp 2018

Fundamentals of Mathematics II, Westmont College

Sp 2019, Sp 2017

Publications

[In Preparation](#)

- [1] Kyle Hansen. *Volumes of Nullhomotopies in Nilpotent Spaces*. PhD thesis, UCSB, 2025.

[Under Review](#)

- [2] Kyle Hansen. Simple tilings of nilpotent lie groups. *arXiv preprint arXiv:2402.10194*, 2024.

[Published](#)

- [3] David Hansen and Kyle Hansen. Clues about bluffing in Clue: Is conventional wisdom wise? *IEEE Transactions on Games*, 13(3):310–314, 2019.
- [4] Yury Ionin and Kyle Hansen. A special instance of rock–paper–scissors. *The American Mathematical Monthly*, 125(8):757–758, 2018.

Selected Presentations

- 2025 **Who Cut the High-Dimensional Cheese?**, *Grad Slam*, UCSB, Santa Barbara, CA
- Mathematical Quasicrystals: The Taming of the Hullabaloo**, *Colloquium*, Wheaton College, Wheaton, IL
- 2024 **Tiling Spaces Which are Fiber Bundles over Nilmanifolds**, *41st Workshop in Geometric Topology*, Calvin University, Grand Rapids, MI
- 2022 **The Hull of a Manifold with Bounded Geometry**, *Topology Seminar*, UCSB, Santa Barbara, CA
- 2019 **When the Fundamental Theorem of Algebra Goes Awry**, *Contributed Paper Session on Revitalizing Complex Analysis*, JMM, Baltimore, MD
- Joint with Mike Brilleslyper (Florida Poly) and Russ Howell (Westmont College)

Research Interests

[Graduate Research](#)

- 2022-2025 **Quantitative Topology, UCSB**, Santa Barbara, CA
- Quantify the volumes of nullhomotopies in nilpotent spaces
- Advisor: Fedya Manin

Tiling Spaces, UCSB, Santa Barbara, CA

Determine topological structures of tiling spaces generated by tilings of nilpotent Lie groups

Advisor: Fedya Manin

Nilpotent-Elliptic Manifolds, UCSB, Santa Barbara, CA

Detect nilpotent-ellipticity (in the sense of Gromov) of a manifold from its algebraic data

Advisor: Fedya Manin

Undergraduate Research

2018-2019 **Complex Analysis, Westmont College, Santa Barbara, CA**

Count and locate the roots of complex harmonic polynomials

Advisor: Russell Howell

2017 **Computational Graph Theory, National Security Internship Program at the Pacific Northwest National Lab, Richland, WA**

Detect scale-invariant properties of large social networks

Advisor: Jennifer Webster

Mentorship, Volunteer, and Community Service

2024 **JMM Chaperone, Westmont College, Santa Barbara, CA**

Drive college students to the Joint Mathematics Meeting, and debrief conference talks

2024-2025 **Graduate Christian Encouragement (GraCE) Coordinator, Unaffiliated, Santa Barbara, CA**

Organize a weekly informal gathering of Christian graduate students at UCSB

2023-2025 **Directed Reading Program Mentor, UCSB, Santa Barbara, CA**

Mentor undergraduates in elective topics: tiling spaces, cryptography, quantum computing

2022-2024 **Math Field Day Volunteer, Westmont College, Santa Barbara, CA**

Guide undergraduate volunteers in judging high school "chalk talks" in annual competition

2022-2025 **Community Group Discussion Leader, Christ Presbyterian Church, Santa Barbara, CA**

Lead adults in church community group discussions

2022-Present **Youth Group Volunteer, Christ Presbyterian Church, Santa Barbara, CA**

Mentor high school students in small groups and individually

2022-Present **Quire of Voyces, Santa Barbara City College, Santa Barbara, CA**

Sing in a local classical a capella choir

2020-Present **Church Setup and Security, Christ Presbyterian Church, Santa Barbara, CA**

Prepare church building for Sunday worship, and ensure safety of attendees

2020-2021 **Elementary School Virtual Tutor, Partners in Education, Santa Barbara, CA**

Assist elementary school teachers in overseeing remote math class activities

Conferences & Workshops Attended

2024 **41st Workshop in Geometric Topology, Calvin University, Grand Rapids, MI**

Quantitative Topology and Beyond, Simons Foundation, New York City, NY

Joint Mathematics Meeting, San Francisco, CA

2023 **Geometry and Analysis of Groups and Manifolds, Centro De Giorgi, Pisa, Italy**

Sub-Riemannian Geometry and Beyond III, Centro De Giorgi, Pisa, Italy

2022 **Graduate Student Concentration Week on Metric Geometry, Texas A & M Workshop in Analysis and Probability, College Station, TX**

2019 **Joint Mathematics Meeting, Baltimore, MD**

Honors & Awards

2024 Department of Mathematics Outstanding Teaching Assistant Award

UCSB

2019 Outstanding Graduate Award in Mathematics
2019 Kenneth Monroe Award

Westmont College
Westmont College

Technical Skills

Programming

LaTeX, Python, R, SQL, Java, Ruby, C, Racket (Scheme)

Teaching Tools

Canvas, Gradescope, WeBWork, Piazza, iClicker